Mitchel Carson

Machine Learning Engineer & Applied AI Researcher · Active TS/SCI

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SUMMARY

M.S. Artificial Intelligence student at UT Austin (4.0 GPA, expected May 2027) doing hydrology-ML research: reforecast generation software for NOAA's NextGen framework and a deep-learning post-processor that improves streamflow forecasts at 1–18 hour lead times, comparing LSTM, vanilla Transformer, and Mamba-style state-space models under leakage-aware evaluation. Accepted AGU26 scientific workshop facilitator. Production GraphQL services at USAA; U.S. Air Force veteran with cloud research experience on Google Cloud and AWS.

RESEARCH AND SELECTED PROJECTS

HYDRA · NextGen reforecast generation and post-processing research — 2025 – present

○ IN PROGRESS · MANUSCRIPTS IN PREPARATION FOR WATER RESOURCES RESEARCH AND ENVIRONMENTAL MODELLING & SOFTWARE

- Built NextGen reforecast generation software that drives NOAA's NextGen framework to produce retrospective streamflow forecasts, plus tooling that acquires, verifies, and tidies the data (public: github.com/Mitchel34/NextGen_Hydra).
- Developing a deep-learning post-processor to improve NextGen forecasts at 1–18 hour lead times; comparing LSTM, vanilla Transformer, and Mamba-style state-space models (with attention) on identical inputs and splits, evaluated by site and lead time (RMSE, NSE, KGE).
- Acquire hydrologic forecast data with Google Cloud BigQuery and Cloud Storage, preserving initialization, lead, valid-time, version, and source metadata; leakage-aware temporal splits follow forecast availability.
- Preparing a results manuscript for *Water Resources Research* and a software paper for *Environmental Modelling & Software*.

Harmony · AI-agent and finance research system on AWS — 2025 – present

○ ACTIVE DEVELOPMENT

- Building a modular Python research system that separates data input, forecasting, validation, policy controls, audit state, and simulation; fail-closed checks stop workflows when data or evaluation rules are incomplete.
- Designed an isolated AWS research-compute path for reproducible CPU/GPU experiments while keeping local tooling as the control and review plane.

CONFERENCE ACTIVITIES

● ACCEPTED Scientific workshop facilitator, *Best Practices for AI and Agentic Workflows in Earth Science Research*, AGU26 Annual Meeting, San Francisco. — Dec 7–11, 2026

◐ UNDER REVIEW HYDRA streamflow-forecasting abstract, AGU26 session H100 (machine learning in hydrology). — Decision pending

PROFESSIONAL EXPERIENCE

USAA · Software Engineering Intern · Global Headquarters, San Antonio, TX — May – Aug 2025

- Developed GraphQL API capabilities in Java and Spring Boot for customer-data workflows used across enterprise channels.
- Enhanced an internal troubleshooting tool by integrating source-of-record and non-source-of-record name and employment data; built JavaScript comparison views for business users.
- Delivered changes through an Agile/Scrum workflow with Git and Jira, in collaboration with product owners, engineers, and internal users.

United States Air Force · Executive Missions Aviator · JB Andrews, MD — Aug 2020 – Apr 2023

- Maintained passenger safety, logistics, and schedule reliability for distinguished guests aboard Air Force 2.
- Coordinated mission planning with flight crews, security teams, and executive staff under exacting operational standards.

EDUCATION

M.S. Artificial Intelligence — Expected May 2027

The University of Texas at Austin · GPA 4.0/4.0
Completed: AI Ethics, Machine Learning, Deep Learning, Reinforcement Learning. Fall 2026: Advances in Deep Learning; Optimization; Natural Language Processing.

B.S. Computer Science — Dec 2025

Appalachian State University, Boone, NC · Cum Laude · GPA 3.6/4.0
Data Science Certificate. Senior Honors Thesis on runoff forecasting with deep learning.

TECHNICAL SKILLS

LANGUAGES	Python, Java, JavaScript/TypeScript, SQL, R
MACHINE LEARNING	PyTorch; LSTM, Transformer, and Mamba-style state-space sequence models; forecast post-processing; leakage-aware temporal evaluation; hydrologic metrics (RMSE, NSE, KGE); pandas, NumPy, xarray
CLOUD	Google Cloud (BigQuery, Cloud Storage) for hydrologic forecast data; AWS research compute (isolated CPU/GPU environments) for AI-agent and finance research; Docker
SOFTWARE	Spring Boot, GraphQL, React, Node.js, PostgreSQL, Git, Jira, Agile/Scrum